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RESEARCH

The Association Between the Impact of COVID-19 and Internalizing Problems Among Children and Adolescents with ADHD: The Moderating Role of Parental Anxiety

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Abstract

Previous research has identified an increase in internalizing problems during the COVID-19 pandemic in children and adolescents with attention-deficit/hyperactivity disorder (ADHD). Additionally, it has been observed that parents of children with ADHD had elevated levels of anxiety during the pandemic. The current study aimed to longitudinally assess whether the impact of COVID-19 was associated with internalizing problems in children and adolescents with ADHD during the middle (Time 1–Spring 2021 [T1]) and end (Time 2–Fall/Winter 2022 [T2]) of the pandemic, and whether parental anxiety moderated this relationship over time. Canadian parents of youth with ADHD (aged 3–18 years old) completed online questionnaires assessing their child's depression and anxiety symptoms, their own anxiety symptoms, and the pandemic's impact on their child, both at T1 ($N = 278$) and T2 ($N = 89$). The results indicated that the impact of COVID-19 on children at T1 was a unique predictor of child internalizing problems at T1 but not at T2. While parental anxiety did not moderate this association cross-sectionally, it was a significant moderator longitudinally. More specifically, low parental anxiety at T1 positively moderated the association between the COVID-19 impact on children at T1 and child internalizing problems at T2. The results highlight the importance of providing on-going psychological support for children and adolescents with ADHD and emphasize the need to aid parents in effectively supporting their children during the process of pandemic recovery.

Keywords ADHD · COVID-19 · Child and adolescent · Internalizing problems · Parental anxiety

Introduction

The 2019 coronavirus (COVID-19) pandemic, and associated public health measures, disrupted the lives of families, children, and adolescents worldwide since its emergence. Many researchers have raised concerns about pandemic-related increases in internalizing problems in both parents and children [1–3]. Internalizing problems are symptoms of negative emotions turned toward the self, such as anxiety and depression [4]. Research shows that children and adolescents with attention-deficit/hyperactivity disorder (ADHD), a neurodevelopmental disorder (NDD) characterized by impairing symptoms of inattention, and/or hyperactivity and impulsivity [5], appear to be more at risk for various negative mental health symptoms due to pandemic-related changes in their lives [6]. Although the COVID-19 pandemic is no longer classified as a global health emergency, there remains a need for research to identify and understand the ongoing effects of

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the pandemic on children's mental health. Thus, this longitudinal study offers valuable insights into the enduring internalizing problems faced by children and adolescents with ADHD during and after the COVID-19 pandemic, while considering the role of parent's own anxiety in this context.

The impacts of the COVID-19 pandemic on children and adolescents have been widely documented. Such impacts include, for example, changes to, and lack of, a daily routine [7], decreased amounts and quality of sleep [8], increased screen time [9], and feeling socially disconnected [10]. The COVID-19 pandemic was exceptionally challenging for individuals with NDDs such as ADHD. Compared to their neurotypical peers, children and adolescents with ADHD experienced more learning difficulties [11], greater sleep-related challenges [12], and spent more time on a screen [13]. Disruptions to structured routines, like those resulting from the pandemic, have a significant effect on young individuals with ADHD. It has been previously documented that the lack of a structured and stable routine for children with ADHD can lead to elevated symptoms of internalizing problems like anxiety, depression, as well as other mental health consequences such as elevated ADHD symptoms (Currie et al., [6, 14]).

It is widely known that individuals with ADHD are particularly susceptible to other mental health concerns such as internalizing problems, even prior to the pandemic [15]. Unfortunately, ADHD often coincides with other mental health disorders, with research indicating that as many as 50% of children and adolescents diagnosed with ADHD also experience concurrent anxiety or depression [16]. Furthermore, studies show that children and adolescents diagnosed with ADHD who have comorbid internalizing problems are at increased risk for other difficulties such as sleep problems [17], Lycett et al., [18], loneliness [19], and having worse therapeutic treatment outcomes [20].

An overwhelming number of studies have found an increase in child and adolescent internalizing problems throughout the COVID-19 pandemic [21], Barbieri et al. [21, 22]. This increase has been linked to factors such as the loss of social activities, lockdown/public health restrictions, unstable routines, and the perceived negative impact of the pandemic on one's life [23–25]. Considering that many of these pandemic impacts are marked by disruption to routine and instability, it is unsurprising that those with ADHD, who are typically more sensitive to change, have been identified as more vulnerable to internalizing problems during the pandemic [3]. Compared to their neurotypical peers, children and adolescents with ADHD experienced greater increases in internalizing problems throughout various timepoints during the pandemic [26–28]. The increased susceptibility to adverse mental health consequences in this population underscores the necessity to further research and understand their pandemic-related internalizing problems and assess whether these problems continue to persist.

Prior to and during the COVID-19 pandemic, internalizing problems have been associated with many contextual and relational factors for children and adolescents with ADHD, with one factor being parental anxiety. Parental anxiety is characterized by experiences of excessive worry, fear, and anxiousness [29]. During the pandemic, Johnson et al. [2] found that 25% of parents experienced anxiety symptoms, while Khomaeny and Kusumaputeri [30] found that over 63% of parents reported anxiety symptoms. Research even suggests that individuals experienced greater increases in anxiety over other mental health symptoms, such as depression, during the pandemic [31], Muñoz-Navarro et al., 2020. In addition, studies have found that parents of children with ADHD experienced greater anxiety symptoms during the pandemic compared to parents of children without ADHD (Chafouleas et al. [32, 33]. This heightened parental anxiety has been related to the worsening of children's own anxiety symptoms during the pandemic [34]. Anxiety also took a unique role in the pandemic as it was somewhat normalized and deemed "adaptive" as it often resulted in abiding by COVID-19 restrictions and lowering chances of infection (Gul Deniz [35]). Therefore, it seems anxiety played a large role in the lives of parents during the pandemic and this anxiety has been related to their own child's anxiety as well as their child's ADHD.

Pandemic research exploring parental mental health in relation to child internalizing problems has been explored in neurotypical populations. Frigerio et al. [36] observed that higher levels of maternal negative mood symptoms were linked to an increase in child internalizing problems from before the pandemic to during the pandemic. Likewise, Lengua et al. [37] found increases in maternal anxiety to be associated with increases in adolescent internalizing problems at two pandemic timepoints, with this relationship being mediated by pandemic-related stressors [37]. Khoury and colleagues' (2021) Canadian study found that compared to pre-pandemic reports, children's internalizing symptoms increased during the pandemic, with maternal anxiety being associated to this increase.

Further research is needed to examine the interplay between COVID-19, child internalizing problems, and parental anxiety among children and adolescents with ADHD. Although this relationship has been explored among those without ADHD, and this research provides a basis for our understanding on this relationship among those with ADHD, a focus must be placed on children and adolescents with ADHD considering their sensitivity to change and elevated vulnerability to worsening internalizing problems Chafouleas et al. [6, 32, 33]. Moreover, since many studies have been conducted within an evolving pandemic context, little is known about how children and adolescents

are adapting after the pandemic ceased to be a global health emergency, as research in this area is in its infancy. It is likely that the pandemic may still have enduring effects on children's mental health, considering how major, stressful life events have been known to be associated with child internalizing problems even many years after they occur (e.g., [38–40]). Therefore, continued research is needed to examine the longitudinal association between the impact of the COVID-19 pandemic, child internalizing problems, and parental anxiety among young individuals with ADHD.

The Current Study

The central aim of the present study is to examine how the impact of COVID-19 is related to internalizing problems in children and adolescents with ADHD during and after the pandemic and whether parental anxiety during the pandemic moderated this association. The findings will provide insight into the lasting effects of the pandemic and inform service providers and parents on how to better support children with ADHD in mitigating negative, pandemic-related mental health consequences. It is hypothesized that the impact COVID-19 had on children and adolescents with ADHD will be positively related to their internalizing problems at both timepoints and that parental anxiety will moderate this association. To clarify, at both Time 1 (T1) and Time 2 (T2), it is hypothesized that high levels of parental anxiety will lead to increasing child internalizing problems as the impact of COVID-19 also increases. These hypotheses are based on prior research with neurotypical children which suggests that high maternal mental health symptoms predict greater increases in child internalizing problems during the pandemic compared to before the pandemic [36, 37, 41]. The study hypotheses are also supported by research that indicates that children with ADHD are more vulnerable to change (Currie et al., [6, 14]), experiencing internalizing problems [26–28], and having a parent with elevated anxiety [6, 32, 33] during the pandemic.

Method

Procedure

The present study was part of a larger, on-going longitudinal survey study examining the experiences of Canadian parents and their school-aged children (with and without ADHD) throughout the pandemic. The current study used two timepoints: T1 (Spring 2021) and T2 (Fall/Winter 2022; originally Time 3 as part of the larger study, but referred to as T2 for the purposes of this study). All methods and procedures used in this study were approved by the Human Research Ethics Board of the University of Ottawa and the Carleton University Research Ethics Board. Participants were recruited through participating universities research labs, professional platforms, and social media platforms. Participant responses were obtained using the Qualtrics Survey Platform, where consenting parents completed a series of questionnaires at the two timepoints. Only parents who initially agreed to receive email follow-ups at T1 were contacted for participation at T2. The survey took approximately 30 min to complete, and participants received a $20 gift card at each timepoint for their participation.

Participants

An extensive data filtering procedure was applied at both timepoints (i.e., excluding participants who were outside of Canada, completed the survey in less than 10 min, who's child was not between 3 and 18 years old, and removing duplicate responses). Participants from T2 were matched to T1 and only children with a parent-reported formal ADHD diagnosis from a clinician were included in the final samples. This data filtering process to remove ineligible participants and bot responses resulted in the use of 23.2% of original responses at T1 and 58.6% of original responses at T2. The final sample at T1 consisted of 278 parents (60.7% biological mothers) and their child with ADHD (59.0% males, $M_{age} = 11.9$ years, $SD = 2.9$). At T2, the final sample consisted of 89 of the same parents (68.5% being biological mothers) and their same child (53.9% males, $M_{age} = 12.6$ years, $SD = 2.7$). Detailed demographic information is presented in Table 1.

Considering this sizeable attrition rate, descriptive analyses were conducted to assess whether participants who only participated at T1 and chose not to continue to T2 differed significantly compared to those who participated in both T1 and T2. These analyses revealed that parents who did not participate in T2 worked more hours per week ($t = -4.51$, $p < 0.001$), had lower income ($t = 2.95$, $p = 0.004$), and were more likely to have a child with another mental health diagnosis ($t = 4.45$, $p < 0.001$). In addition, parents who only participated at T1 reported lower anxiety symptoms ($t = -7.12$, $p < 0.001$) and less symptoms related to ADHD ($t = -2.04$, $p = 0.04$). The parents who dropped out also reported that their child had fewer internalizing symptoms ($t = -4.96$, $p < 0.001$) and was less impacted by the pandemic ($t = -6.00$, $p < 0.001$).

Measures

The Impact of COVID-19 on Children and Adolescents

To determine the level of disruption children and adolescents experienced due to the COVID-19 pandemic, a modified version of the *Co-SPACE COVID-19 Impact*

Table 1 Sample demographic characteristics

Characteristic	Time 1		Time 2	
	n	%	*n*	%
Child age (in years)				
6	3	1.2	1	1.1
7	10	4.1	2	2.2
8	28	11.5	6	6.7
9	14	5.7	2	2.2
10	26	10.7	8	9.0
11	26	10.7	8	9.0
12	35	14.3	12	13.5
13	31	12.7	14	15.7
14	28	11.5	16	18.0
15	13	5.3	8	9.0
16	9	3.7	6	6.7
17	14	5.7	3	3.4
18	7	2.9	3	3.4
Child sex				
Male	144	59.0	48	53.9
Female	100	41.0	41	46.1
Child ethnicity				
Aboriginal/First Nations/Métis/Inuit	53	21.7	8	9.0
White/Caucasian	151	61.9	72	80.9
South Asian	1	0.4	1	1.1
Black	12	4.9	3	3.4
Latin American	3	1.2	–	–
Japanese	1	0.4	1	1.1
Multi-racial	2	0.8	–	–
Other	1	0.4	–	–
Child co-morbid disorders				
Attachment disorder	29	11.9	7	7.9
Adjustment disorder	32	13.1	10	11.2
Communication disorder	48	19.7	21	23.6
Depression	33	13.5	22	24.7
Anxiety disorders	2	0.8	–	–
Obsessive compulsive disorder	38	15.6	17	19.1
Bipolar disorder	44	18.0	21	23.6
Tourette syndrome	9	3.7	5	5.6
Conduct disorder	6	2.5	2	2.2
Oppositional defiant disorder	10	4.1	2	2.2
Autism spectrum disorder (e.g., Asperger's Syndrome, Autism, PDD-NOS)	8	3.3	1	1.1
Other	3	1.2	–	–
Province/Territory				
Alberta	39	16.0	15	16.9
British Columbia	42	17.2	21	23.6
Manitoba	14	5.7	5	5.6
New Brunswick	8	3.3	2	2.2
Newfoundland and Labrador	11	4.5	2	2.2
Nova Scotia	75	4.1	3	3.4
Ontario	10	30.7	32	36.0
Quebec	27	11.1	3	3.4
Saskatchewan	4	1.6	1	1.1

Table 1 (continued)

Characteristic	Time 1		Time 2	
	n	%	*n*	%
Northwest Territories	9	3.7	2	2.2
Nunavut	3	1.2	2	2.2
Annual family income				
Lower class ($7000–$32,999)	17	6.9	2	2.2
Lower-middle class ($33,000–$52,999)	34	13.9	15	16.8
Middle class ($53,000–$105,999)	124	50.7	58	65.1
Middle-upper class ($106,000–$264,999)	57	23.4	11	12.4
Upper class ($265,000 +)	8	3.2	3	3.4
Number of hours parents work weekly				
0–5	2	0.8	–	–
6–20	31	12.7	2	2.2
21–39	116	47.5	38	42.7
40 or more	95	38.9	49	55.1
Number of adults in the home who care for the children				
1	16	6.6	6	6.7
2	154	63.1	60	67.4
3	51	20.9	21	23.6
4	13	5.3	2	2.2
5	9	3.7	–	–
6	1	0.4	–	–
Primary language spoken in home				
English	241	98.8	87	97.8
French	2	0.8	2	2.2
Spanish	1	0.4	–	–
Parent relationship to child				
Biological mother	148	60.7	61	68.5
Biological father	89	36.5	27	30.3
Stepmother	2	0.8	–	–
Stepfather	2	0.8	1	1.1
Foster mother	2	0.8	–	–
Adoptive father	1	0.4	–	–

$N = 244$ for Time 1 and $N = 89$ at Time 2. Time 2 valued endorsed by Time 1

Scale [42] was completed by parents. This measure aims to gain a deeper understanding of how families coped with certain changes and consequences of the pandemic. The measure was modified to fit the parameters and goals of the current study. The modified scale consisted of 9 items, rated on a 4-point Likert scale, ranging from 1 (*not at all)* to 4 (*completely).* Parents assessed the frequency at which their child encountered changes in various aspects of their life (i.e., daytime routine, bedtime routine, diet, exercise, screen time, stress, anxiety, socialization, other) due to the COVID-19 pandemic over the past week (e.g., "increased screen time"). The scale showed acceptable internal consistency at T1 ($\alpha = 0.79$). This variable is referred to in-text as "COVID-19 child impact".

Child Internalizing Problems

Parents completed the *Child and Adolescents Symptom Inventory-Progress Monitor-Parent Version* (*CASI-PM-P*; [43]) to assess their child's internalizing symptoms. This measure is a comprehensive 29-item scale that is typically administered multiple times over a certain period of time as it is used to assess long-term change in symptoms related to mental health disorders in children and adolescents, upon referral to mental health facilities/services. These disorders include generalized anxiety disorder (GAD), social phobia, separation anxiety disorder, major depressive disorder (MDD), ADHD and its subtypes, oppositional defiant disorder (ODD), and conduct disorder (CD). Parents filled out

all 29 items, however, in the current study, only 13 items from the various anxiety subscales and depression subscale were used to compute an internalizing score considering their internalized nature. More specifically, the four items from the GAD subscale (e.g., "Is extremely tense or unable to relax"), the two items from the social phobia subscale (e.g., "Is excessively shy with peers"), the three items from the separation anxiety subscale (e.g., "Gets very upset when child expects to be separated from home or parents"), and the four items from the MDD subscale (e.g., "Is depressed/sad for most of the day") were totaled to create an internalizing score. Parents indicated how often their child exhibited each behaviour in the past month on a 4-point scale from 1 (*never*) to 4 (*very often*). The internalizing scale proved to have good internal consistency at T1 ($\alpha = 0.87$) and T2 ($\alpha = 0.81$).

Parental Anxiety

Parental anxiety was assessed through the anxiety subscale of the *Emotional Distress Scale* which is part of the Patient-Reported Outcomes Measurement Information System [44]. *The Emotional Distress Scale* is an items bank that consists of 28 depression-related items, 29 anxiety-related items, and 29 anger-related items. However, a modified version of this scale was administered such that parents filled out a four-item anxiety subscale and four-item depression subscale. The current study focused on the four-item anxiety subscale (e.g., "I felt uneasy"). Parents were asked to rate how much each statement applied to them in the past 7 days using a 5-point Likert scale from 1 (*never*) to 5 (*always*). In the current sample, the four-item anxiety subscale demonstrated good internal consistency at T1 ($\alpha = 0.80$) but not at T2 ($\alpha = 0.48$), suggesting a potential limitation when applied to this population. However, prior research in similar populations reported good internal consistency ($\alpha = 0.89$) for this scale as well as strong validity [45].

Results

Descriptive statistics for the primary study variables, which include COVID-19 child impact (T1), child internalizing problems (T1, T2), and parental anxiety (T1, T2), are presented in Table 2. Child age was not significantly associated to child internalizing problems at T1 ($r = -0.06$, $p = 0.31$) nor T2 ($r = -0.10$, $p = 0.38$) and thus was not used as a covariate in further analyses. All regression analyses performed with these variables were tested against the appropriate assumptions. The assumptions of linearity, homoscedasticity, normality, multicollinearity, and undue influence were satisfied.

Table 2 Descriptive statistics of study variables

Variable	*n*	*M*	*SD*	Range	
				Min	Max
COVID-19 child impact (T1)	275	2.32	0.52	1.00	4.00
Child internalizing problems (T1)	275	1.36	0.53	0.00	3.00
Parental anxiety (T1)	277	2.63	0.82	1.00	5.00
Child internalizing problems (T2)	89	1.35	0.45	0.08	2.33
Parental anxiety (T2)	88	2.73	0.53	1.50	4.25

Main Analyses

Regression and Moderation Analyses at Time 1

A simple linear regression was conducted to test the hypothesis that COVID-19 child impact would positively predict children's internalizing problems at T1. Overall, the results provided support for the hypothesis. COVID-19 child impact (T1) was a significant predictor of internalizing problems (T1) ($R^2 = 0.25$, F (1, 271) $= 89.58$, $p < 0.001$; $b = 0.50$, $SE = 0.53$, $p < 0.001$, $\eta_p^2 = 0.50$, CI 95% [0.40, 0.61]). To assess the moderating role of parental anxiety (T1) on the relation above, a simple moderation analysis, as well as all subsequent moderation analyses and post-hoc analyses, was conducted using the "PROCESS" macro, v. 4.2 (which also mean centred all variables prior to analyses) [46]. The overall model was statistically significant, $R^2 = 0.42$, F (3, 269) $= 65.99$, $p < 0.001$. Yet, contrary to the hypothesis, parental anxiety was not a significant moderator of the relationship between COVID-19 child impact and child internalizing problems at T1 (see Table 3).

Regression and Moderation Analyses at Time 2

Next, a simple linear regression was run to assess whether COVID-19 child impact at T1 longitudinally predicted child internalizing problems at T2. The overall model was not significant, $R^2 = 0.03$, F (1, 87) $= 3.02$, $p = 0.09$, such that the COVID-19 child impact from spring 2021 did not significantly predict internalizing problems in Fall/Winter 2022, contrary to our hypothesis. Further, a simple moderation analysis was conducted to test the moderating role of parental anxiety at T1 between COVID-19 child impact at T1 and child internalizing problems at T2. The overall model was significant, $R^2 = 0.20$, F (3, 85) $= 6.92$, $p < 0.001$. Moreover, a significant and negative interaction effect was identified, providing confirmation that parental anxiety at T1 indeed moderated the relationship between COVID-19 child impact at T1 and child internalizing problems at T2 (see Table 4).

A simple slopes analysis was conducted to determine the nature of the moderating effect of parental anxiety. The effect levels of parental anxiety were mean centered during

Table 3 Moderation model of predictors of child internalizing problems at Time 1

	b	*SE*	*t*	*p*	95% *CI*	
					Lower	*Upper*
Constant	1.35	0.03	51.45	< .001	1.30	1.41
COVID-19 child impact (centered)	0.22	0.06	3.98	< .001	0.11	0.33
Parental anxiety (centered)	0.32	0.04	8.94	< .001	0.25	0.39
COVID-19 child impact X parental anxiety	0.00	0.04	−0.03	0.97	−0.08	0.08

$N = 273$

Table 4 Moderation model of predictors of child internalizing problems at Time 2: parental anxiety at Time 1

	b	*SE*	*t*	*p*	95% *CI*	
					Lower	*Upper*
Constant	1.39	0.04	31.25	< .001	1.31	1.48
COVID-19 child impact (centered)	0.01	0.14	0.09	0.93	−0.26	0.29
Parental anxiety (centered)	0.17	0.07	2.28	0.03	0.02	0.31
COVID-19 child impact parental anxiety	−0.54	0.15	−3.71	< .001	−0.83	−0.25

$N = 89$

Table 5 Simple slopes analysis examining the relation between COVID-19 child impact at Time 1 and child internalizing problems at Time 2 at different levels of parental anxiety at Time 1

	b	*SE*	*t*	*p*	95% *CI*	
					Lower	*Upper*
Low parental anxiety (-1SD)	0.36	0.15	2.47	0.02	0.07	0.65
Average parental anxiety (mean)	0.01	0.14	0.09	0.93	−0.26	0.29
High parental anxiety (+1SD)	−0.33	0.18	−1.81	0.07	−0.70	0.03

$N = 89$

analysis and levels of the moderator were determined as follows: low (1 SD below the mean), average (mean), and high (1 SD above the mean). As seen in Table 5, the findings indicated that COVID-19 child impact at T1 positively predicted child internalizing problems at T2, but only at low levels of parental anxiety (T1) (see Fig. 1). Thus, these results did not provide support for the study's hypothesis that high parental anxiety would act as a moderator in the relationship above.

A final simple moderation was conducted to assess the hypothesis that parental anxiety at T2 moderated the relation between COVID-19 child impact at T1 and child internalizing problems at T2. The overall model was significant, $R^2 = 0.17$, $F\ (3, 84) = 5.86$, $p = 0.001$, with a significant, positive main effect of parental anxiety. As seen in Table 6, no significant interaction was observed, indicating that parental anxiety at T2 did not serve as a moderator in the relationship between COVID-19 child impact at T1 and child internalizing problems at T2.

Lastly, post-hoc power analyses were run using G*Power [47] to assure the moderation analyses at T2 were adequately powered given the modest sample size. The post-hoc power analysis for the moderation testing the moderating role of parental anxiety at T1 between COVID-19 child impact at T1 and child internalizing problems at T2 revealed an achieved power of 0.98 ($N = 89$, $\alpha = 0.05$, $f^2 = 0.24$). Similarly, the results from the post-hoc power analysis for the moderation testing the moderating role of parental anxiety at T2 between COVID-19 child impact at T1 and child internalizing problems at T2 revealed an achieved power of 0.96

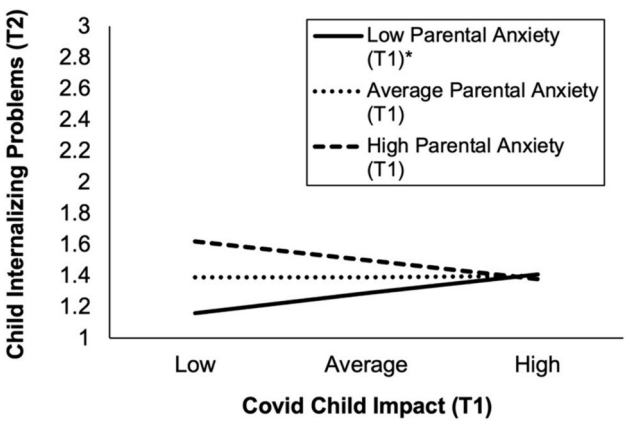


Fig. 1 Graph of the simple slopes analysis of the moderating effect of parental anxiety at Time 1 in the relation betweenCOVID-19 child impact at Time 1 and child internalizing problems at Time 2. *$p < 0.05$

Table 6 Moderation model of predictors of child internalizing problems at Time 2: parental anxiety at Time 2

	b	*SE*	*t*	*p*	95% *CI*	
					Lower	*Upper*
Constant	1.35	0.04	30.20	< .001	1.26	1.44
COVID-19 child Impact (centered)	0.12	0.14	0.92	0.36	−0.14	0.39
Parental anxiety (centered)	0.33	0.09	3.79	< .001	0.16	0.50
COVID-19 child Impact X parental anxiety	−0.03	0.21	−0.15	0.88	−0.44	0.38

$N = 88$

($N = 88$, $\alpha = 0.05$, $f^2 = 0.21$). Thus, these tests confirm that the performed analyses were well-powered and accurate at detecting any true effects.

Discussion

This study investigated the longitudinal association between the impact of COVID-19 on children and adolescents with ADHD and their internalizing problems, while analyzing the moderating role of parental anxiety. This study specifically examined children and adolescents with ADHD due to their vulnerability to routine changes and high rate of comorbid internalizing, mental health disorders (Chafouleas et al. [32]; Currie et al. [14, 28]). The present study aimed to provide novel insight and understanding regarding the internalizing problems of Canadian youth with ADHD during and towards the end of the pandemic.

In accordance with the current pandemic literature on the mental health of children with ADHD [3, 23, 27], the results from this study indicated that this population experienced worsening anxiety and depression symptoms as the pandemic increasingly affected their lives, regardless of parental anxiety. However, low parental anxiety during the pandemic was a significant contributing factor to prolonged elevation of internalizing problems among children and adolescents who were greatly impacted by the pandemic. This longitudinal finding contradicts findings of past pandemic research on neurotypical youth, which often indicates an association between high levels of negative parental mental health symptoms (such as anxiety) and high child internalizing problems [36, 37, 41]. While previous pandemic research has established a link between parental mental health and child mental health in neurotypical children both cross-sectionally [37] and longitudinally [36, 41], the current study uniquely uncovered a delayed impact of parental anxiety on COVID-19-related child internalizing problems in children and adolescents with ADHD. To explain further, in children and adolescents with ADHD, being greatly impacted by the COVID-19 pandemic was related to elevated anxiety and depression symptoms a year later (when the pandemic was no longer a global health emergency), but only when the child's parent had minimal anxiety. This finding, as well as literature suggesting stressful life events impact children's internalizing symptoms several years after their occurrence [38–40], underscore the importance of investigating this relationship over time, even beyond the peak of the COVID-19 global health crisis as this effect seems to be long-lasting. This finding also emphasizes the need for further ADHD-specific research considering this population's unique vulnerabilities and potentially unique pandemic-related outcomes.

There are different explanations as to why the moderating effect of low parental anxiety was discovered in this study, as opposed to the predicted moderating effect of high parental anxiety. Firstly, one possibility for the found moderation model is a lack of parental emotional support. During the pandemic, parents lack of encouragement for their child to express their emotions has been related to worse child anxiety [48]. Therefore, as potentially seen in the present study, a parent with low anxiety, who does not experience the same level of internalizing problems as their child, may struggle to express empathetic support due to a lack of lived experience, potentially exacerbating their child's internalizing problems. Alternatively, it is also well-understood that parent internalizing problems, such as anxiety, are positively related to child internalizing symptoms [49, 50]. It is therefore possible that in this study high parental anxiety so strongly elevates child internalizing problems that other factors, such as the impact of COVID-19, do not cause much significant variability in a child's internalizing problems as they have already reached a peak. Yet, in relation to this study, this possibility does not account for the lack of a significant increase in internalizing problems in children of parents with average levels of anxiety.

Furthermore, the results derived from the significant moderation model in this study suggest that lower parental anxiety could act as a protective factor in minimizing the development of internalizing problems in children and adolescents with ADHD. As depicted in Fig. 1, during low impact, child internalizing problems closely mirrored levels of parental anxiety such that children's internalizing problems were lower when parents had low anxiety levels. This aligns with aforementioned, pre-pandemic research that indicates a positive relationship between parent and child mental health in which when parents exhibit lower levels of

anxiety, their children also tend to experience lower levels of anxiety (Adolph et al., 2020; [51]). Although this finding suggests a protective effect of lower parental anxiety, this effect appears to be limited to low impact situations as depicted in this study.

Implications

The study findings suggest that Canadian children and adolescents with ADHD experienced adverse emotional effects as a result of the COVID-19 pandemic and that these challenges have persisted beyond the pandemic's peak. In response to a novel crisis such as the pandemic, it is recommended that public health measures prioritize maintaining pre-pandemic practices (i.e., daily living and routines) as this approach could help to minimize the potential harm to mental health among vulnerable youth. Moreover, it is imperative that mental health resources, psychological support, and coping strategies, become widely accessible to children and adolescents with ADHD both during and after a pandemic as the impact of the COVID-19 pandemic on these young people's lives has shown to be associated with poorer internalizing outcomes.

Furthermore, the results of this study shed light on the importance of parents when it comes to their children's internalizing problems. Treating parental anxiety, even when relatively mild, with therapeutic and parenting exercises has been shown to reduce impairment in children with ADHD [52]. Additionally, as discussed by Davidson and colleagues (2022), strong parental mental health literacy should be sought in collaboration with parents and health care providers as it better allows parents to recognize their child's mental health challenges and connect them to support. Moreover, parents could be encouraged to participate in family-based cognitive behavioural therapy with their child as this has shown to decrease internalizing symptoms in children and adolescents with ADHD [53].

Limitations

This study has several limitations to consider. In this sample, the internal consistency of the *Emotional Distress Scale* [44] was low at T2, which was not a concern at T1, nor consistent with prior research [45]. Although it is thus advisable to approach the results with some caution due to the relatively lower reliability, this outcome is expected. Variability in constructs has been found in longitudinal analyses of the anxiety subscale from the *Emotional Distress Scale* [54] as well as in similar measures [55, 56]. However, this variability has shown to have small practical differences in scores [54]. Another significant limitation in the current study was that it was limited to the report of a singular parent and therefore the model is unable to account for the possible impact of both parents' anxiety. Relatedly, having a single informant potentially allows for common method variance. Furthermore, although children and adolescents have been shown to have a better assessment of their own internalizing symptoms compared to their parents [57], children and adolescents with ADHD have been found to be inconsistent in self-reporting (Hoza et al. 2002). Future studies using multi-informant data in the form of dual-parent participation as well as child/self-report may clarify the associations between the variables in this study and avoid potential biases.

Lastly, there was an attrition rate of 68% from T1 to T2. Further examination of those who dropped out showed that parents who only participated in T1 worked significantly more hours but had a lower overall income than those who participated at both timepoints. Moreover, parents who dropped out had less anxiety and less symptoms related to ADHD themselves. Also, they had a child with fewer internalizing problems and who was less impacted by the pandemic. Thus, it is possible that individuals of this sub-sample that dropped out had minimal interest in further participating due to time restraints and priorities (i.e., work, money), and/or less investment as they were not seeing much impairment in their child and might not understand the impacts of ADHD considering they have lesser related symptoms themselves. Although, future research may want to clarify the longitudinal relationship examined in this study among this sub-sample of parents who dropped out after T1, all main study analyses were re-run with family income and parental work hours as covariates which yielded no significant differences to study findings.

Future Directions

In addition to addressing the limitations of this study, future research should consider the inclusion of other parental mental health variables as potential moderators. For instance, factors like parental ADHD, which has been linked to child internalizing problems and child ADHD (Humphreys et al., 2010), should be considered for further investigation. Furthermore, it would be beneficial to replicate this study among different populations of children, including those of diverse backgrounds (e.g., low SES, considering the large attrition rate in this study among those of lower income but who work more hours) or those with NDDs other than ADHD (e.g., autism spectrum disorder), as these populations also seem to be more vulnerable to adverse mental health outcomes due to the pandemic [58, 59]. Finally, future research should continue to explore the underlying mechanisms and factors that may have contributed to the significant moderation model uncovered in this study. For instance, it would be valuable to empirically examine the effect of parental emotional support given its effect on child

mental health (Maiuolo et al., 2019, Repetti et al., 2002), and its probable value when other forms of social support are lacking during a pandemic.

Conclusion

The present study contributes to the understanding of how the COVID-19 pandemic and parental anxiety influenced the development and persistence of internalizing problems in children and adolescents with ADHD. The study revealed that as the impact of COVID-19 increased, child and adolescent internalizing symptoms also increased, and this relationship was longitudinally moderated by low levels of parental anxiety. This research provides evidence of the significant and negative impact that the COVID-19 pandemic had, and continues to have, on the mental health of young individuals with ADHD. These findings emphasize the need for public health measures to prioritize stability in daily routines and to ensure access to mental health care during and after major national/global events, such as the COVID-19 pandemic. The results also highlight the need to equip parents and children with appropriate mental health resources and strategies to navigate the emotional challenges that arise from a novel global crisis.

Summary

The COVID-19 pandemic was a trying time, with children and adolescents with ADHD showing heightened vulnerability to pandemic-related changes and internalizing problems. Pandemic studies have shown elevated anxiety and depression symptoms in children and adolescents with ADHD to be related to their parents worsening of anxiety. Although the intersection of COVID-19, child internalizing problems, and parental anxiety has been explored in neurotypical populations, it had yet to be explored in children with ADHD, nor in a post-pandemic setting. Therefore, the current longitudinal study assessed whether child internalizing problems were linked to the impact of COVID-19 and whether parental anxiety moderated this relationship during (Time 1–Spring 2021) and after (Time 2–Fall/Winter 2022) the pandemic in a Canadian sample. At Time 1, 278 parents and their child with a formal ADHD diagnosis, participated, and 89 of the same parents and their same child participated at Time 2. At both timepoints, parents filled out measures about the impact of COVID-19 on their child, their child's internalizing problems, and their own anxiety. A series of linear regressions and simple moderations were run. Results revealed that the impact of COVID-19 uniquely predicted child internalizing problems during the pandemic, but not after. Experiencing a high impact of the pandemic on one's life was found to be related to worse internalizing problems a year later, but only among children and adolescents who had a parent with low anxiety. These findings demonstrate the potent role that COVID-19 and parental anxiety played on the mood of children and adolescents with ADHD during the pandemic, and even after it was over. Evidently, it is imperative that continued mental health support and education be given to these young people with ADHD as well as their parents. These findings inform future research and on the lasting effects of the pandemic on vulnerable youth.

Author Contribution Lauren Gowans is the lead author of this paper and conceptualized this specific paper, conducted the analyses, as well as wrote the entire manuscript. Maria A. Rogers and Yuanyuan Jiang are the principal investigators and intellectual leads on the larger, original study. Emma A. Climie, Janet W.T. Mah, and Penny Corkum are co-investigators who advised during all stages of the study. Amanda Krause and Tessa Ritchie played an advisory role throughout the project. Marina Parvanova contributed to the writing of the document.

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Data Availability Participants in this study did not consent to have their data made public.

Declarations

Conflict of interest The authors declare they have no conflict of interest.

Consent to Participate Participants (parents) gave their informed consent to participate in the study at both T1 and T2 via the Qualtrics survey platform.

Ethics Approval Approval was obtained from the ethics committee of Carleton University and the University of Ottawa. The procedures used in this study adhere to the tenets of the Declaration of Helsinki.

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